THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · EASY LEVEL CHAPTER TEST SET 2

Chapter 6: **How Forces Affect Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. When a bus starts suddenly we move: 1

(a) forward **(b)** upward **(c)** backward **(d)** sideways

2. A 2 kg body moving at 5 m/s has momentum: 1

(a) 2.5 kg m/s **(b)** 25 kg m/s **(c)** 7 kg m/s **(d)** 10 kg m/s

3. F = 1

(a) a/m **(b)** m + a **(c)** ma **(d)** m/a

4. When a bus stops suddenly we move: 1

(a) backward **(b)** forward **(c)** upward **(d)** sideways

5. Net force on a body with balanced forces: 1

(a) infinite **(b)** positive **(c)** negative **(d)** zero

6. Momentum = 1

(a) mv **(b)** m/v **(c)** v/m **(d)** m + v

7. Wheels reduce: 1

(a) mass **(b)** weight **(c)** speed **(d)** friction

8. An unbalanced force causes: 1

(a) rest only **(b)** no change **(c)** constant speed **(d)** acceleration

9. Newton's third law concerns: 1

(a) gravity **(b)** inertia

(c) action and reaction **(d)** energy

10. Oil reduces: 1

(a) mass **(b)** friction **(c)** gravity **(d)** weight

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Define the SI unit of force. 2

12. Find the force for m = 4 kg and a = 2 m/s squared. 2

13. State Newton's third law. 2

14. What is friction? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Give three examples of Newton's third law. 3

16. Find the weight of a 50 kg person on Earth (g = 10). 3

17. A force of 6 N acts on a body of mass 2 kg. Find the acceleration and the velocity after 3 s from rest. 3

OR

State three ways of reducing friction. 3

18. State three differences between mass and weight. 3

OR

Why do we use lubricants in machines? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Derive F = ma. 5

OR

A car of mass 800 kg accelerates from rest to 20 m/s in 10 s. Find the acceleration and force. 5

20. A force of 40 N acts on a 8 kg body. Find a, v after 5 s and s. 5

OR

Explain balanced and unbalanced forces with examples. 5

END OF QUESTION PAPER — Total: 40 Marks

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